USN

[22CSE124]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2023

FUNDAMENTALS OF MACHINE LEARNING

Time: 3 Hours Maximum **Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	a.	What is Machine learning?	L-1	CO1	**02**
	b.	Difference between independent and dependent variable?	L-1	CO2	**02**
	c.	Elements of reinforcement learning?	L-1	CO3	**02**
	d.	Differentiate between Deep Learning and Machine Learning.	L-1	CO4	**02**
	e.	What is semi- supervised learning?			
		PART-B	**06 X 05=30**		
2.	a.	Explain optimization in ML?	L2	CO1	**05**
		OR			
	b.	Explain different ways of data preprocessing?	L2	CO1	**05**
3.	a.	Difference between supervised and unsupervised learning?	L2	CO2	**05**
		OR			
	b.	What is Exploratory Data Analysis?	L2	CO2	**05**
4.	a.	Explain gradient descent algorithm?	L2	CO3	**05**
		OR			
	b.	Define feature selection and Feature extraction?	L2	CO3	**05**
5.	a.	What are supervised learning algorithms? Explain any one in detail.	L2	CO4	**05**
		OR			
	b.	Differentiate Recurrent Neural Networks from feed forward neural networks?	L2	CO4	**05**
6.	a.	What is unsupervised learning algorithm? Name any two of unsupervised learning algorithms?	L2	CO5	**05**
		OR			
	b.	Mention the importance of Pandas usage in Machine learning.	L2	CO5	**05**
7.	a.	Difference between Unsupervised and semi supervised learning?	L2	CO6	**05**
		OR			
	b.	Explain exploding gradient and diminishing gradient ?	L2	CO6	**05**

		PART-C	**06 X 10=60**		
8.	**a.**	Explain SVM in detail with diagram?	L2	CO1	**10**
		OR			
	b.	What is role of activation function in Neural Networks ? how is it important ?	L2	CO1	**10**
9.	**a.**	What is principle component analysis? How do we decide Pc1 and Pc2 while calculating the new dimensions?	L2	CO2	**10**
		OR			
	b.	What is a cost function in Neural networks? Name any two coat functions in ML?	L2	CO2	**10**
10.	**a.**	Explain delta rule with an example.	L2	CO3	**10**
		OR			
	b.	What is feed-forward and back propagation in ML?	L2	CO3	**10**
11.	**a.**	What is bias in Machine learning? How can it be ethically mitigate bias in ML?	L2	CO4	**10**
		OR			
	b.	Difference between variance and covariance?	L2	CO4	**10**
12.	**a.**	Explain linear regression and logistic regression?	L2	CO5	**10**
		OR			
	b.	Mention the importance of perceptron.	L2	CO5	**10**
13.	**a.**	What is dimensionality reduction and how to calculate Eigen vectors and Eigen Values?	L2	CO1	**10**
		OR			
	b.	What is the main purpose of CNN?	L2	CO1	**10**